From Motherhood Penalty to Caregiver Penalty: How Large Language Model Résumé Screeners Treat Family Signals

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Abstract

Large language models (LLMs) now screen job applicants. We ask whether they reproduce the motherhood penalty and, if so, what they penalize. On a résumé, motherhood is a bundle of three signals that human hiring cannot pull apart: the applicant is a woman, she has family responsibilities, and she has often been out of work. We unbundle these signals in an audit of four widely used models (616,419 decisions, 28 occupations), adding and removing each one from otherwise identical résumés, removing all gender information, and observing four decision margins: scoring one applicant, choosing between two, shortlisting from a pool, and declining to decide. The models do not penalize being a woman. They penalize caregiving. The penalty is invisible when applicants are scored one at a time but large when they are compared: a mother returning from parental leave wins a head-to-head choice against an equally qualified childless woman 11 to 31 percent of the time. It is unchanged when gender information is removed, falls on fathers as much as on mothers, and consists of a penalty for interrupted employment plus a further penalty for family signals. Gender-specific behavior exists only as a thin layer that differs by vendor. Fairness instructions reduce the interruption penalty but not the family-signal penalty. We interpret the pattern as the ideal-worker norm rendered as a formally gender-neutral rule, which can reproduce a gendered outcome because women carry caregiving signals more often than men.

1 Introduction

Mothers pay a price in the labor market that fathers do not. They earn less than comparable childless women (Budig and England 2001), and in audit and laboratory studies employers rate them as less competent and committed, recommend them for hire less often, and offer them lower salaries, while fathers are not penalized (Correll, Benard, and Paik 2007). The penalty persists when mothers present direct evidence of commitment, which suggests that it rests on norms about what mothers should do as well as on inferences about what they can do (Benard and Correll 2010).

Hiring is now partly delegated to software. Résumé screening is among the most common commercial uses of large language models, and among the most regulated: US selection procedures are judged by their disparate impact on protected groups, and the European Union classifies AI systems used in recruitment as

high-risk (EEOC 1978; European Union 2024). Whether these systems reproduce the motherhood penalty is therefore a pressing question. This paper argues that it is also an opportunity to answer an older one: *what, exactly, is penalized when mothers are penalized?*

On a résumé, motherhood is not a single signal. It bundles three: the applicant is a woman; she has family responsibilities, which applications reveal through mentions of children, parenting activities, or caregiving leave; and she has often been out of work. Each component corresponds to a distinct explanation of the penalty. If employers penalize women as presumed caregivers, the gender component matters. If they penalize anyone who departs from the norm of the fully available "ideal worker" (Acker 1990; Williams 2000), the family and interruption components matter, whoever carries them. In human hiring the components travel together, and they are hard to separate: an employer who sees a mother's name sees her gender, and audits cannot remove gender while keeping the family situation visible. An LLM screener can be shown the same résumé thousands of times with one component added or removed at a time, including with every trace of gender removed. That makes it possible to unbundle the motherhood signal.

Figure 1 lays out the framework that organizes the paper. We treat the motherhood signal as three separable components (left), ask which of them an LLM screener penalizes and whether it needs to know the applicant's gender to do so (center), and observe the result at four decision margins (right): scoring a single applicant, choosing between two finalists, shortlisting from a pool, and declining to decide. Three questions follow. *Where* does a family-status penalty appear? *What* part of the signal is penalized? And *with what consequences*—for remedies that developers and employers rely on, and for women and men as groups?

Our answers are as follows. The penalty appears only when applicants are compared; single-applicant scores barely move. It is a penalty on caregiving rather than on gender: it is unchanged when the model cannot know the applicant's gender, it falls on fathers as much as on mothers, and it has two parts—a penalty for any interruption in employment and a further penalty for overt family signals. Gender-specific behavior does exist, but as a thin layer that differs from vendor to vendor. Fairness instructions reach the interruption penalty but not the family-signal penalty. And because women carry caregiving signals more often than men, a rule that is gender-blind in form can produce a gendered outcome.

The paper makes three contributions. Substantively, it separates components of the motherhood penalty that human hiring confounds and shows which survive in automated screening. Theoretically, it suggests that when gender is the category developers train against, discrimination may persist through correlated signals that are not protected as such, consistent with the ideal-worker account. Methodologically, it shows that audit design determines what is found: audits that score single applicants, discard non-answers, or manipulate gender only through names would certify these systems as fair.

2 Background

2.1 Three components, three accounts

Explanations of the motherhood penalty can be read as claims about which component of the signal does the work (Figure 1, center).

Gender-statistical accounts hold that employers treat motherhood as a proxy for lower expected availabil-

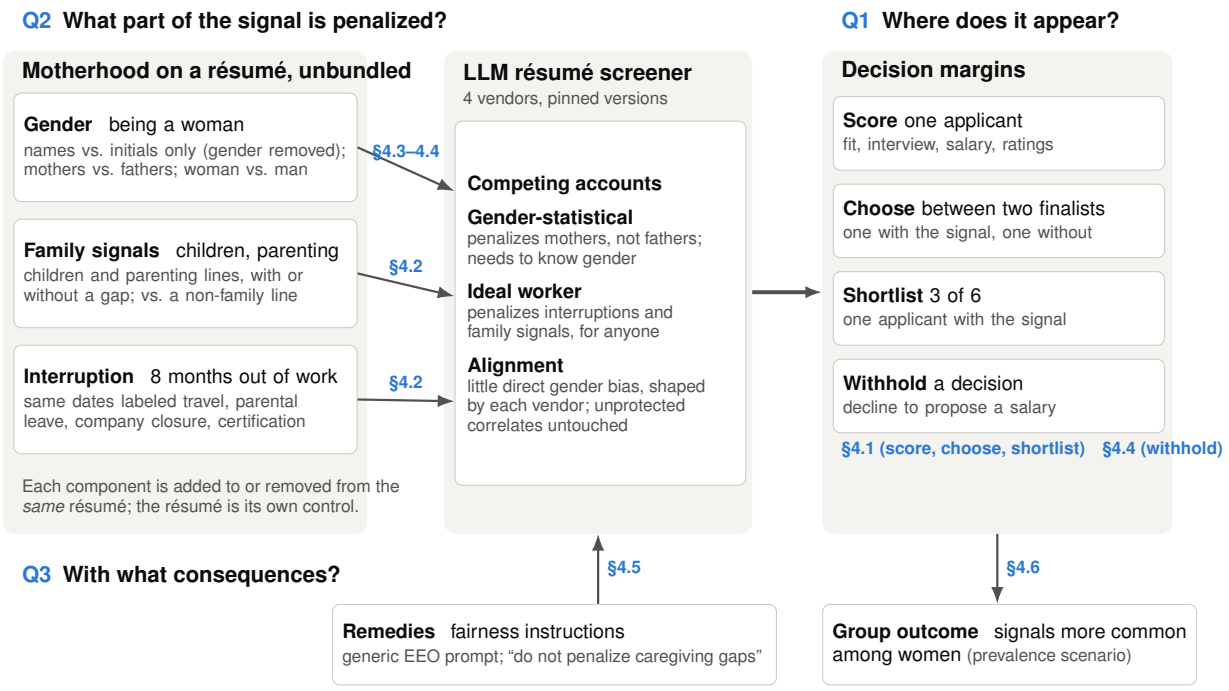


Figure 1: Analytical framework. The motherhood signal is unbundled into three components (left), each added to or removed from the same résumé; competing accounts disagree about which components an LLM screener penalizes (center); the result is observed at four decision margins (right) and traced to remedies and group outcomes (bottom). Section numbers locate each test.

ity and productivity, and that the inference is gendered: mothers, not fathers, are expected to reduce their commitment (Phelps 1972; Correll, Benard, and Paik 2007). A normative variant holds that mothers violate expectations of devotion to family, so that evidence of commitment does not erase the penalty (Benard and Correll 2010). Both predict a penalty specific to mothers, which requires the evaluator to know the applicant's gender.

Ideal-worker accounts locate the penalty in the organization of work. Organizations are built around an abstract worker who is fully available and free of outside obligations (Acker 1990); the norm is gender-neutral in wording but was modeled on men whose care work was done by others (Williams 2000; Blair-Loy 2003). On this view the penalty attaches to signs of competing obligations and to departures from continuous employment, whoever shows them. Two strands of evidence support the distinction. Men who signal caregiving are penalized too (Rudman and Mescher 2013; Williams, Blair-Loy, and Berdahl 2013), and employment gaps are penalized whatever their reason (Kroft, Lange, and Notowidigdo 2013; Weisshaar 2018). Because women take most caregiving leaves (Goldin 2014), such a penalty is gendered in its consequences even when it is not gendered in its application.

An alignment account is specific to the machine setting. Commercial LLMs are trained and evaluated to avoid differential treatment of protected groups, and audits of automated hiring have mostly asked whether outcomes differ by protected attributes signaled through names (An et al. 2025; Armstrong et al. 2024; Rozado 2025; Wilson and Caliskan 2024), following the logic of anti-discrimination law (Barocas and Selbst 2016). Systems may therefore be scrutinized and corrected along the gender dimension while correlated

signals that are not themselves protected go unexamined. This account predicts little direct gender bias, bias that varies with each developer's choices, and untouched penalties on unprotected correlates.

2.2 Expectations

Mapped onto the framework, the accounts make distinguishable predictions. The gender-statistical account predicts that the gender component carries the penalty: mothers are penalized more than fathers, and the penalty shrinks when gender information is removed. The ideal-worker account predicts that the family and interruption components carry it: fathers are penalized as much as mothers, removing gender information changes nothing, and both an unexplained gap and overt family signals are costly. The alignment account predicts a weak and vendor-specific gender component that responds to instructions. The last two can hold together. We state these as orienting expectations; the confirmatory tests specified before data collection are listed in Section 3.5.

The framework also implies where to look. Selection decisions are comparative, and a penalty that reflects relative judgments about availability may surface only when applicants compete. If audits measure only single-applicant scores, they may miss it.

3 Design

The study is a correspondence-style audit (Bertrand and Mullainathan 2004) carried out on LLMs rather than employers. Its design follows Figure 1: one set of manipulations unbundles the signal, a second controls what the model knows about gender, a third varies the decision margin, and a fourth tests remedies. Every estimate compares a résumé with itself, with and without a component, so that applicant quality is held constant by construction.

3.1 Unbundling the signal

Table 1 lists the signals. Every application includes a short cover note, so that conditions differ only in what the note and the résumé say. The *parent* signal adds "married with two children" to the note and a parent-teacher association line to the résumé. *Parent with leave* adds an eight-month "Parental leave" entry to the work history. Three placebo signals separate the components. A *travel gap* inserts an eight-month "Career break—extended travel" with the same dates, isolating the interruption from any family content. *Parental leave only* inserts the leave entry without children or parenting activity, isolating the caregiving label. A *personal* placebo adds a length-matched non-family line, isolating the cost of disclosing anything personal. Comparing these conditions yields four components: the interruption (travel gap against equal treatment), its caregiving label (parental leave only minus travel gap), family signals on top of a labeled leave (parent with leave minus parental leave only), and parenthood without a gap net of disclosure (parent minus personal). Because "extended travel" may itself be stigmatized, a later phase replaces it with two neutral labels for the same dates—"Job search after company closure" and "Full-time certification program." Two further signals probe the normative account: *parent with commitment evidence* adds objective commitment bullets, and a

Table 1: Unbundling the motherhood signal

Signal	What it adds to the same résumé	What the comparison isolates
None	neutral cover note	baseline
Personal (placebo)	a length-matched hobby and volunteer line	disclosure as such
Travel gap (placebo)	8-month "Career break — extended travel"	the interruption (vs. equal treatment)
Parental leave only (placebo)	the same dates labeled "Parental leave"	the caregiving label (vs. travel gap)
Parent	"married with two children"; parent-association line	parenthood without a gap (vs. personal)
Parent + leave	Parent plus the parental-leave entry	family signals on top of a leave (vs. leave only)
Neutral gap labels	same dates as "company closure" or "certification"	whether travel is stigmatized
Parent + commitment	Parent plus objective commitment bullets	the normative account
Pregnant (women only)	"currently pregnant"	pregnancy, incl. withholding

Note: Signals render identically for women and men; every application includes a cover note. Gender information is varied separately (names; name or initial plus an explicit gender line; initials only).

pregnancy signal, for women only, states that the applicant is pregnant.

The signals are deliberately explicit, which favors identification over realism. The parental-leave gap is the most realistic of them and anchors the decomposition.

3.2 Controlling gender information

In the main run, gender is conveyed by first names: six per gender from Bertrand and Mullainathan (2004), all coded as White to hold race constant (Gaddis 2017), rotated within every condition. A second phase varies gender information at four levels: the name alone; the name plus a header line stating "Gender: Female/Male"; a gender-neutral initial plus that line; and initials only, which removes gender entirely. If the penalty is about gender, it should change across these levels. The same phase includes direct choices between a woman and a man with the same family status, which measure gender preference as such. A perception check confirms that the manipulation works. Asked directly, models stated the gender in explicit header lines correctly in every case but often answered "unknown" for names (Luna 34 percent, Haiku 58 percent); yet whenever a model used a gendered pronoun in an explanation, it matched the name or header in every case, and gendered pronouns all but disappeared under initials only. The models read gender from names; the direct question measures their willingness to say so.

3.3 Observing decision margins

Each signal is evaluated at four margins. In *scoring*, the model evaluates one applicant: fit on a ten-point scale, an interview recommendation, a salary, and ratings of competence, commitment, likability, and promotability. In *pairwise choice*, it picks one of two finalists of the same gender, one with the signal and one without, with résumé roles and screen positions counterbalanced; under equal treatment the applicant with the signal is chosen half the time. In *shortlisting*, it selects three of six applicants, one of whom carries the signal; the

outcome is that applicant's shortlisting rate relative to the same résumé without the signal, an impact ratio comparable to the four-fifths rule. *Declining to decide*—withholding a salary offer or refusing to answer—is recorded as an outcome rather than discarded.

3.4 Testing remedies

A final phase asks whether the instructions that deployers typically add remove the penalty. The same pairwise tasks are run under three system prompts: none (the default elsewhere in the study), a generic equal-employment-opportunity prompt, and the same prompt plus "do not penalize employment gaps taken for caregiving." The second prompt names the interruption but not the family signals, which lets us ask whether the two components respond differently. A further analysis translates each model's measured penalty into group outcomes for women and men under different prevalences of caregiving signals.

3.5 Models, materials, and inference

We audit four widely available, low-cost models—DeepSeek V4.1 Flash, Claude Haiku 5.5, GPT-6 Luna, and Qwen 3.8 Flash—pinned to dated versions and single provider endpoints, with extended reasoning disabled, structured output, and no system prompt except in the remedies phase. They stand for high-volume screening configurations; larger models are not tested. The materials comprise 154 résumés validated as qualified for 28 occupations: 20 spanning the joint distribution of occupational female share (4.7 to 97.1 percent) and median wage, and 8 selected to weaken the correlation between female share and care work (from .53 to .32). Résumés have continuous eight-to-twelve-year histories and no gender or family wording. Data were collected in four phases between October 14 and 30, 2026 (616,419 logged calls; total cost $188.52 plus $0.66 for coding model explanations).

Standard errors are clustered at the units of the design: pairwise choices by the unordered pair of résumés compared (40 pairs in the main occupations, 16 in the added ones), with cluster-level degrees of freedom; scores by résumé; shortlists by occupation with résumé and position fixed effects. Effective sample sizes are thus set by résumé pairs, not by calls. We apply the Holm correction within each family of tests, test claims of no difference for equivalence against a smallest effect of interest of five percentage points (two one-sided tests), check few-cluster tests with wild cluster bootstrap p-values, and re-estimate the main tests as linear probability models with controls for position, prompt wording, and name. Responses produced by repair passes after invalid output are excluded from choice analyses; validity was 99 to 100 percent.

The designs and confirmatory tests—penalties by margin, model, and gender; mother–father differences; impact ratios; the placebo decomposition; gender-information contrasts; direct gender choices; withholding; and the remedy and gap-label hypotheses—were specified in analysis plans committed before each paid run. The five-point equivalence bound for the second phase was chosen after its data were collected but before the tests were run; it was specified in advance for the remedies phase. The study was not registered externally. Other analyses are labeled exploratory, and cross-model comparisons are descriptive.

Table 2: Probability that the applicant with the family signal is chosen in pairwise choice (main run)

Signal — applicant	DeepSeek	Haiku	Luna	Qwen
Parent — mother	.357	.233	.297	.387
Parent — father	.349	.276	.294	.398
Parent + commitment — mother	.419	.266	.228	.407
Parent + leave — mother	.310	.110	.186	.230
Parent + leave — father	.308	.130	.184	.256
Ranking: shortlist rate relative to the same résumé without the signal				
Mothers	.71	.25	.78	.75
Fathers	.70	.23	.81	.80

Note: Pairwise choice against an equally qualified childless candidate of the same gender; .50 = equal treatment. Standard errors clustered by résumé pair (40 pairs) give 95% confidence intervals of ±.02–.07; no interval includes .50. Ranking ratios are résumé fixed-effect estimates from pools of six; .80 is the four-fifths threshold.

4 Results

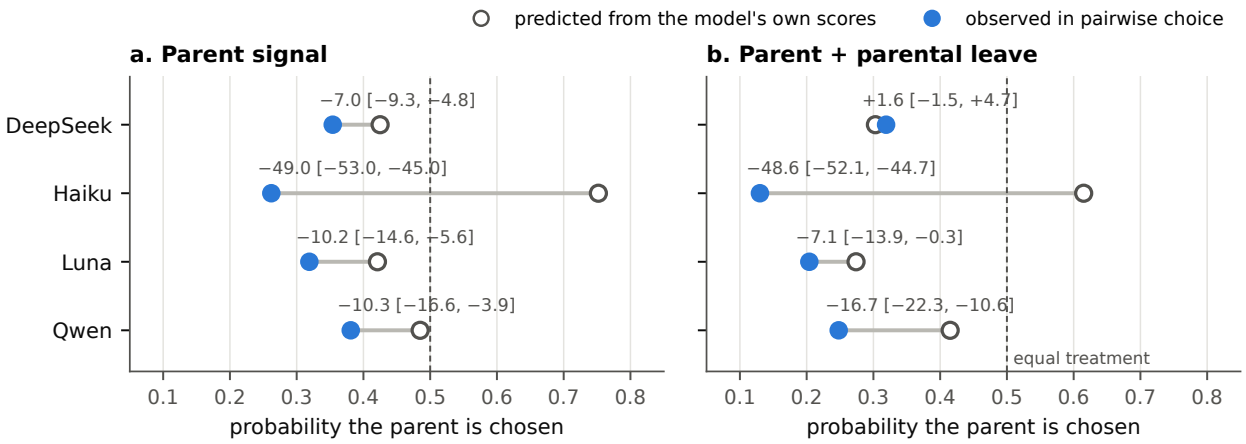
The results follow the three questions of Figure 1. Section 4.1 asks where the penalty appears. Sections 4.2 to 4.4 ask what is penalized: the interruption and family components (4.2), the gender component (4.3), and the gender-specific behaviors that remain (4.4). Sections 4.5 and 4.6 ask about consequences. Differences are in percentage points ("points").

4.1 Where: the penalty appears when applicants are compared

If a screener penalized family signals the way a human rater might, the penalty should be visible in the scores it assigns. It is not. Relative to the same résumé without the signal, the parent signals change women's fit ratings by at most 0.35 points on a ten-point scale and interview recommendations by at most 1.5 percentage points. Claude Haiku even rates mothers 0.34 points *higher* than the same applicants without children.

The same signals are costly when applicants compete (Table 2). In pairwise choice, parents are chosen less than half the time in every model and every condition. A mother returning from parental leave is chosen against an equally qualified childless woman 31 percent of the time by DeepSeek, 23 percent by Qwen, 19 percent by Luna, and 11 percent by Haiku—the model that scores her higher. In shortlisting, the same mothers reach the shortlist at 71 (DeepSeek), 75 (Qwen), and 78 (Luna) percent of the rate they achieve without the signal, each below the four-fifths threshold, and at 25 percent with Haiku.

Comparison does not merely magnify small score differences. Choices respond to ordinary differences in résumé quality at 13 to 18 points of choice probability per fit point; by that yardstick the family signals act like quality losses of 0.9 to 2.0 fit points, many times their effect on scores. A noise model that converts each model's score gap into a predicted choice probability predicts far smaller penalties than observed in seven of eight model-signal combinations (Figure 2); the exception is DeepSeek's leave signal. The penalty is larger when résumés are close in quality, as tie-breaking would imply, but remains 9 to 32 points among pairs with clear quality differences. For Haiku, whose score effect is positive, no magnification of scores can produce



Mothers vs. an equally qualified childless woman. Labels: observed − predicted, percentage points [95% bootstrap CI over résumé pairs]. Prediction converts each model's score gap into a choice probability using its within-résumé rating noise.

Figure 2: Where: scores do not account for choices. Hollow markers: probability that a mother is chosen over an equally qualified childless woman, predicted from the model's own scores; filled markers: observed.

Table 3: Placebo decomposition of the parent-with-leave penalty (percentage points)

Component (comparison)	DeepSeek	Haiku	Luna	Qwen
Interruption	−10.5	−18.4	−26.0	−22.4
(travel gap vs. .50)	($t = -8.0$)	($t = -8.5$)	($t = -7.3$)	($t = -8.9$)
Caregiving label	+1.7	−1.1	+6.5	+6.7
(parental leave only − travel gap)	[+0.5, +2.9]	[−3.2, +0.9]	[+3.7, +9.2]	[+4.9, +8.5]
Family signals on top of leave	−10.6	−17.4	−11.5	−12.1
(parent + leave − parental leave only)	[−12.7, −8.6]	[−20.2, −14.5]	[−15.4, −7.5]	[−14.7, −9.5]
Parenthood net of disclosure	−10.4	−19.2	−12.1	−7.3
(parent − personal placebo)	[−12.8, −8.0]	[−23.4, −15.0]	[−16.3, −7.9]	[−10.3, −4.3]

Note: Mothers; 20 main occupations; gender stated explicitly (levels L1 and L2 pooled); 40 résumé pairs. Brackets: 95% confidence intervals clustered by résumé pair. The personal placebo itself costs 2.9–3.5 points for DeepSeek, Luna, and Qwen and raises Haiku's choice share by 3.7 points.

the observed choice penalty.

The first answer, then, is that the penalty lives at the comparative margins where selection actually happens. The remaining analyses measure it there.

4.2 What: interruptions and family signals

Table 3 and Figure 3 decompose the parent-with-leave penalty in pairwise choice, for mothers in the main occupations with gender stated explicitly (40 résumé pairs). The decomposition answers the ideal-worker account's two claims in turn.

Interruption. An eight-month gap labeled as travel, with no family content, costs 10.5 (DeepSeek) to 26.0 points (Luna). Calling the same gap parental leave adds nothing: the label is statistically equivalent to zero for Haiku and is 1.7 to 6.7 points *less* costly than travel for the other three models. Travel was not an unusually stigmatized reference. When it was replaced by company closure or certification, no neutral

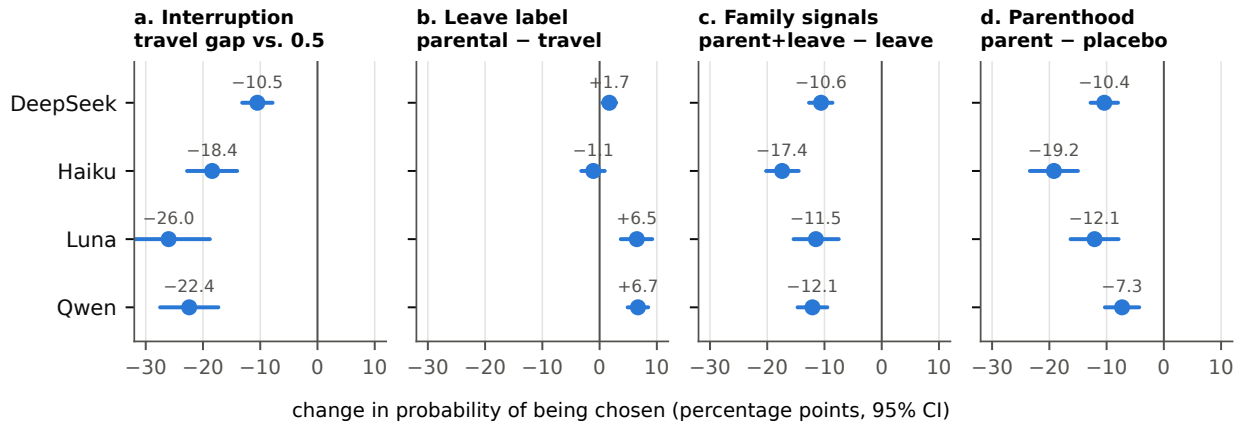


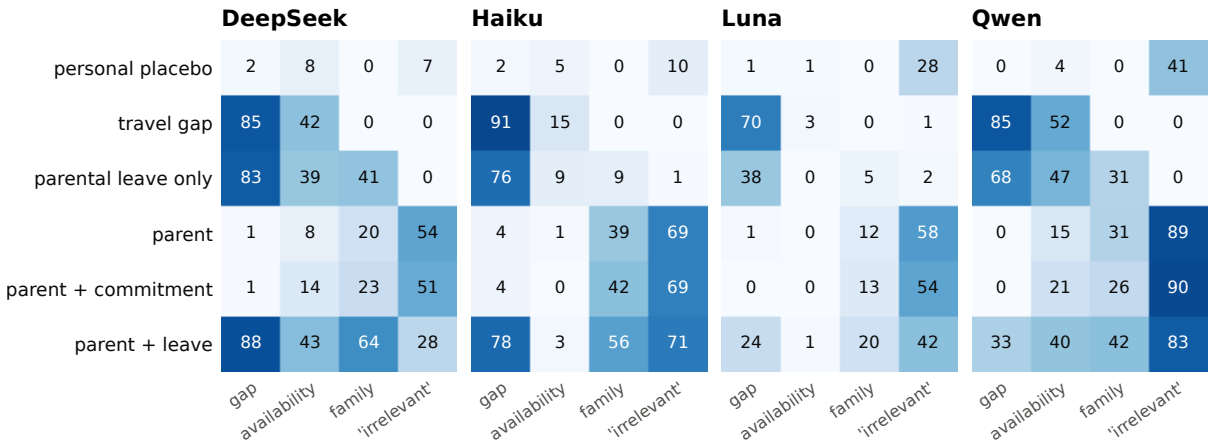
Figure 3: What: decomposition of the parent-with-leave penalty into interruption, caregiving label, family signals, and parenthood without a gap.

label was significantly more costly than travel after correction; Qwen penalized travel 4.3 to 5.4 points more than either neutral label, and seven of the other twelve comparisons were statistically equivalent (the largest estimate in the other direction was DeepSeek's certification label for women, 4.0 points, Holm p = .14). What is penalized is time out of work, not its stated reason.

Family signals. Adding children and parenting activity to a résumé that already carries a labeled parental leave costs a further 10.6 to 17.4 points, and parenthood without any gap, net of the personal placebo, costs 7.3 to 19.2 points. Disclosure as such is not the issue: the placebo costs about 3 points in three models and slightly helps applicants with Haiku (3.7 points). Commitment evidence does not restore equal treatment: it raises mothers' choice share modestly in three models and lowers it in Luna (Table 2), in line with the normative account of the human penalty. The family-signal and parenthood contrasts survive Holm correction in 23 of 24 tests each across six alternative samples of occupations and gender information.

Norm salience. If the interruption penalty reflects an availability norm, making the norm salient should deepen it. Job descriptions with "greedy" availability language ("must be available after hours and on weekends as needed, and travel up to 40% of the time") add to the leave penalty for DeepSeek mothers (2.7 points) and, combined with family-friendly language, for DeepSeek applicants of both genders and Haiku mothers (5.2 to 8.0 points); family-friendly language alone never significantly helps.

What the models say (exploratory). Model explanations follow the decomposition. An LLM coder (DeepSeek V4.1 Flash, temperature zero) classified the reasons given in pairwise-choice explanations; a second coder (Qwen 3.8 Flash) re-coded 638 explanations not written by Qwen. Six codes reached acceptable agreement (κ = .71 to .93); agreement was lower for availability (.68) and family (.63) on explanations not written by DeepSeek. When the applicant with the signal lost, the gap was cited in 24 to 91 percent of explanations for the three gap signals and in at most 4 percent for parent signals without a gap; family was cited in 12 to 64 percent for parent signals and never for the travel gap or the personal placebo (Figure 4). All twelve pre-stated contrasts went in the predicted direction and survived Holm correction. Availability concerns were common for DeepSeek and Qwen (39 to 52 percent of gap-signal losses) but rare for Haiku and



Percent of explanations citing each reason when the applicant with the signal lost; 'irrelevant' = calls family information irrelevant (LLM-coded, exploratory; κ for availability and family is .68/.63 on non-DeepSeek explanations). Darker = more often cited.

Figure 4: What the models say: reasons cited in explanations when the applicant with the signal lost (exploratory).

Luna (15 percent or less). Explanations for losing parents often called family information irrelevant (Qwen 90 percent, Haiku 69, Luna 56, DeepSeek 53); for DeepSeek and Qwen such statements were concentrated in losses (19 and 42 percent when the parent won), for Haiku and Luna they were about as common either way (64 and 50 percent). These are stated reasons, conditional on the outcome and coded without human validation; they describe justifications, not causes.

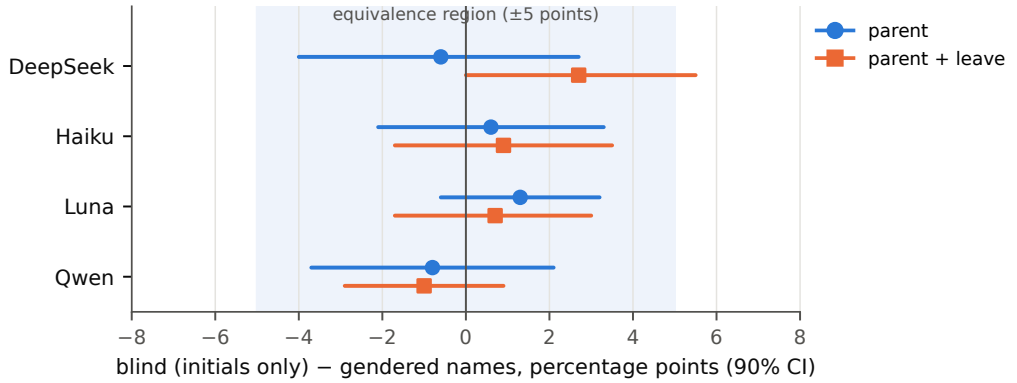
4.3 What: the gender component

The gender-statistical account makes two predictions: the penalty should shrink when the model cannot know the applicant's gender, and mothers should be penalized more than fathers. Neither holds.

Removing gender. In the added occupations, the parent penalty with gendered names differs from the penalty under initials only by +0.6 (DeepSeek), −0.5 (Haiku), −1.3 (Luna), and +0.3 points (Qwen), and the leave penalty likewise. In seven of eight model-signal combinations the difference is statistically equivalent to zero within ±5 points; in the eighth, DeepSeek's leave signal, the interval reaches 5.6 points (Figure 5). The design could detect differences of 3.0 to 5.4 points. Shortlisting replicates the result: ratios barely move across the four levels of gender information (for example, .64/.66/.66/.65 for DeepSeek and .15/.19/.17/.16 for Haiku).

Fathers. Fathers are chosen below equal treatment in every cell of every model; a father returning from parental leave is chosen 31 (DeepSeek), 26 (Qwen), 18 (Luna), and 13 percent (Haiku) of the time against a childless man (Table 2). There is no fatherhood bonus. Mother–father differences on the same résumé pairs are small and, in three models, not significant after correction (Table 4). Haiku is the exception: it chooses mothers 4.4 points less often than fathers (Holm p = .02) and rates them lower on commitment, and its mother–father gap differs significantly from each of the other models.

The penalty, in short, does not require the model to know that the applicant is a woman. It falls on caregivers.



Change in the parent's probability of being chosen when all gender information is removed (8 added occupations, 16 résumé pairs). A CI inside the band means the difference is statistically equivalent to zero.

Figure 5: The gender component: removing all gender information leaves the penalty unchanged.

Table 4: Mother–father differences for the parent signal (main run)

Outcome	DeepSeek	Haiku	Luna	Qwen
Pairwise choice (points)	+0.6	**−4.4**	−0.4	+0.1
	($t = 0.6$)	($t = -3.2$)	($t = -0.4$)	($t = 0.0$)
Paired salary ($)	+138	−150	−109	−16
Commitment rating, paired	+.02	**−.14**	+.00	+.12
Commitment rating, single applicant	**−.09**	**−.06**	+.03	+.00

Note: Mother minus father on the same résumé pairs (choice, paired ratings) or résumés (single-applicant ratings). Bold: significant after Holm correction within outcome.

4.4 The gender layer: thin and vendor-specific

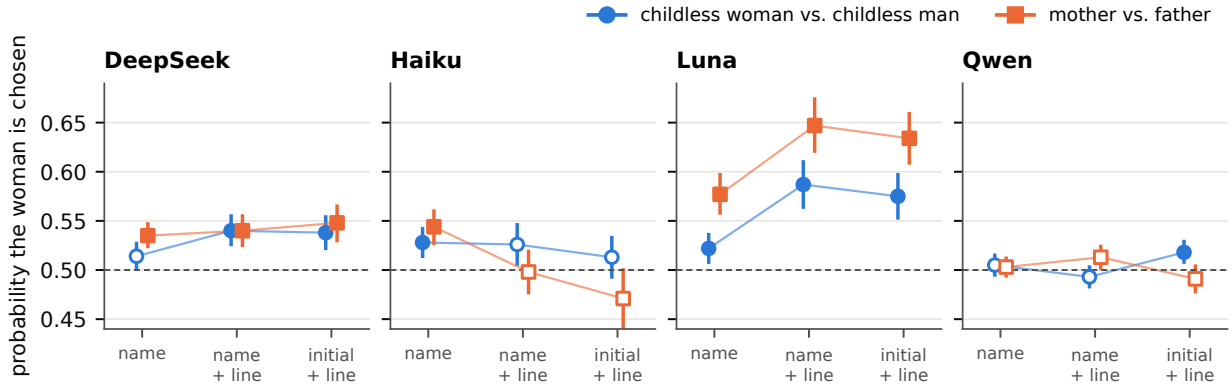
What remains of gender is a set of behaviors that differ by vendor and surface mainly when gender is made explicit. In direct choices between a woman and a man with the same family status (Table 5, Figure 6), Qwen stays within five points of parity at every level, as equivalence tests confirm (its largest deviation is a 1.8-point tilt toward childless women under initials). DeepSeek shows a small preference for women (.535 to .548 for mothers). Luna's preference for women is modest under names (.522 childless, .577 parents) and grows when gender is stated explicitly (.587 and .647). Haiku moves the other way: its slight preference for mothers under names (.544) disappears with an explicit gender line (.498; .471 with initials). Explicit gender fields are therefore not neutral; they activate preferences that differ in direction across vendors.

One vendor's gender layer also appears at the fourth margin. Asked for a salary, Haiku proposes none for 19.8 percent of pregnant applicants in the main run—against 3.1 percent of childless women and 7.3 percent of married women—and for 27.5 percent when gender is stated explicitly; other models withhold for pregnancy far less often (at most 5.8 percent) (Figure 7). Haiku's explanations frame withholding as protective. Its consequence is not: pregnant applicants without an offer are recommended for interview 29.9 percent of the time, against every time when an offer is made, a net interview penalty of 9.4 points relative to childless women. Among the offers that are made, no model shows a meaningful pregnancy salary penalty. An audit that drops missing salaries would conclude that pregnancy is not penalized.

Table 5: Direct woman-versus-man choices: probability the woman is chosen

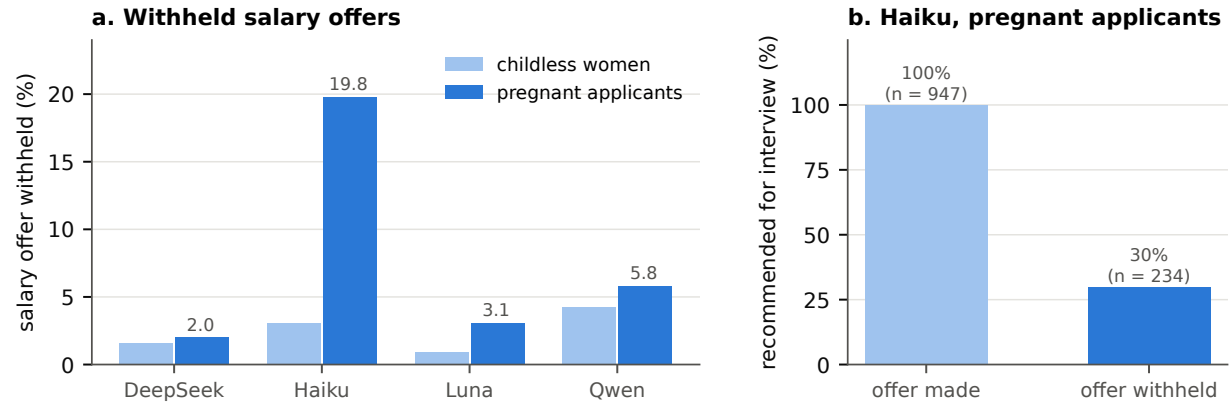
Model	Pair	Names (L0)	Name + line (L1)	Initial + line (L2)
Qwen	childless / parents	.505 / .503	.493 / .513	**.518** / .491
DeepSeek	childless / parents	.514 / **.535**	**.540** / **.540**	**.538** / **.548**
Luna	childless / parents	**.522** / **.577**	**.587** / **.647**	**.575** / **.634**
Haiku	childless / parents	**.528** / **.544**	.526 / .498	.513 / .471

Note: Both candidates share a family status; .50 = no preference; 56 résumé pairs. Bold: differs from .50 after Holm correction (six tests per model). Qwen is statistically equivalent to .50 within ±5 points in every cell.



Both candidates share a family status (56 résumé pairs). 0.5 = no gender preference; above = women preferred. Filled marker = differs from 0.5 after Holm correction; bars = 90% CI. "line" = explicit "Gender: Female/Male" line.

Figure 6: The gender layer: direct woman-versus-man choices by gender information.



Main run, gender conveyed by name. Withholding is a refusal to propose any salary. With an explicit gender line, Haiku withholds for 27.5% of pregnant applicants.

Figure 7: The gender layer at the fourth margin: salary offers withheld (a) and interview recommendations for Haiku's pregnant applicants by whether an offer was made (b).

Table 6: Fairness instructions (phase 4): probability that a mother is chosen, by system prompt

Model	Parent	Parent + leave	Travel gap
DeepSeek	.361 / .325 / .369	.317 / .301 / .386	.422 / .420 / .451
Haiku	.266 / .332 / .342	.074 / .195 / .217	.296 / .308 / .300
Luna	.307 / .357 / .376	.195 / .188 / .286	.238 / .283 / .307
Qwen	.381 / .333 / .351	.234 / .181 / .279	.274 / .286 / .339

Note: Entries: no system prompt / generic equal-opportunity prompt / generic prompt plus "do not penalize employment gaps taken for caregiving." Women; 20 main occupations; 40 résumé pairs; 48,000 calls; .50 = equal treatment. Hypotheses and tests were specified before the run.

4.5 Consequences: remedies reach one component, not the other

Deployers often add fairness instructions to a screener's system prompt. Table 6 reports the remedies phase (48,000 calls). The generic equal-opportunity prompt left the caregiver penalty in place in every model, and its effect depended on the vendor: none for DeepSeek, a reduction of the plain parent penalty for Luna (5.0 points) but not of the leave penalty, substantial reductions for Haiku (6.6 and 12.2 points), and a *deepening* of both penalties for Qwen (4.7 and 5.3 points). The targeted instruction not to penalize caregiving gaps reduced the interruption penalty by 4 to 14 points in all four models but left the family-signal penalty essentially unchanged except in Haiku. No arm brought any model to equal treatment. The gender layer responded more readily: Luna's preference for mothers over fathers fell from .640 to .557 under the generic prompt.

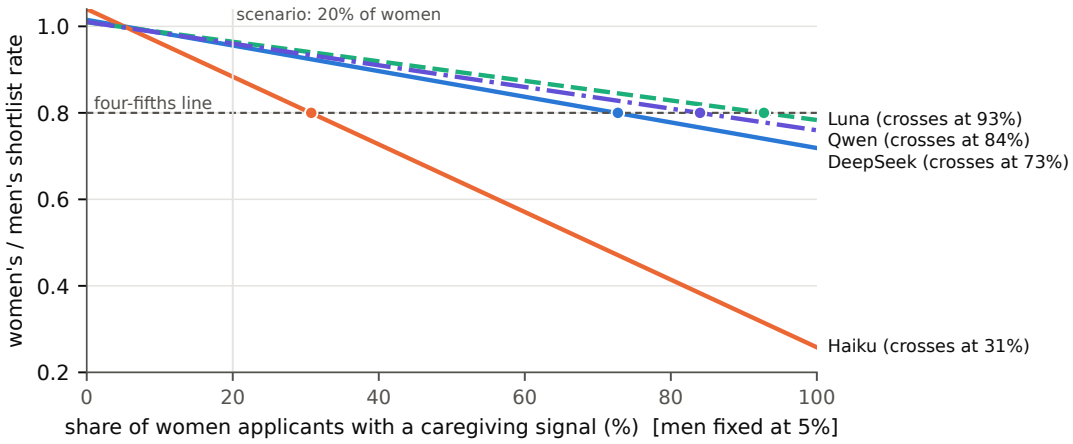
The two components of the caregiver penalty thus behave differently. The interruption penalty, which resembles an inference about availability, responds when an instruction names it. The family-signal penalty, which no instruction names and which resembles a norm, does not.

4.6 Consequences: a gender-blind rule, a gendered outcome

The caregiver penalty is blind to gender, but caregiving signals are not evenly distributed. Applying each model's measured shortlisting penalty to applicant pools in which more women than men carry a caregiving signal shows how the two combine (Figure 8). With men's prevalence fixed at 5 percent, Haiku produces an aggregate women-to-men shortlisting ratio below the four-fifths line once more than 31 percent of women carry the signal; the thresholds are 73 percent for DeepSeek, 84 for Qwen, and 93 for Luna. At a prevalence of 20 percent among women, no model crosses the line (ratios .88 to .96). We could not source gender-specific estimates of how many applicants carry such signals, so prevalence is a scenario parameter; the thresholds do not depend on it. The analysis illustrates the mechanism and shows how much the answer depends on the vendor.

4.7 Additional analyses

In paired salary recommendations, models offer mothers 1.7 to 2.3 percent less than otherwise identical childless women (1.2 to 4.1 percent after a leave)—smaller than the roughly 7 percent gap among human evaluators in Correll, Benard, and Paik (2007), although the designs are not comparable. Across 28 occupa-



Applies each model's measured shortlist penalty, which is the same for women and men, to pools in which women carry the signal more often. Below 0.8 = adverse impact under the four-fifths rule.

Figure 8: Consequences: implied ratio of women's to men's shortlist rates as women's prevalence of caregiving signals rises (men fixed at 5%).

tions, the penalty shows no robust association with occupational female share, wages, or task composition. Linear probability models with controls reproduce the main tests (leave coefficients $-.047$ to $-.161$); wild cluster bootstrap p-values change no conclusion; estimates on a repeated anchor condition drifted by 0.8 to 2.4 points between phases, within the pre-specified tolerance; and screen-position preferences are balanced by design.

5 Discussion

5.1 The framework revisited

Figure 9 returns to the framework with the findings in place. Of the three components of the motherhood signal, the two that the ideal-worker account emphasizes—time out of work and visible family responsibility—carry the penalty, for women and men alike. The component the gender-statistical account emphasizes—being a woman—carries little, and what it carries differs by vendor, as the alignment account would lead one to expect. The penalty appears at the comparative margins where selection happens and is invisible in single-applicant scores.

We therefore describe the screeners' behavior as two layers. The first is a caregiver penalty, shared by all four models in sign, with two parts. The *interruption penalty* behaves like an inference about availability (Phelps 1972; Kroft, Lange, and Notowidigdo 2013; Weisshaar 2018): any eight-month gap is costly whatever it is called, availability language in a job ad deepens it, and an instruction that names it reduces it. The *family-signal penalty* behaves like a norm. It attaches to children and parenting activity, persists on top of a gap already labeled as parental leave, is only partly offset by evidence of commitment, and does not respond to instructions. In these respects it resembles the normative component of the human motherhood penalty (Benard and Correll 2010), except that it is applied to fathers too. The second layer consists of gender-specific behaviors that appear mainly when gender is made salient and differ in direction across vendors.

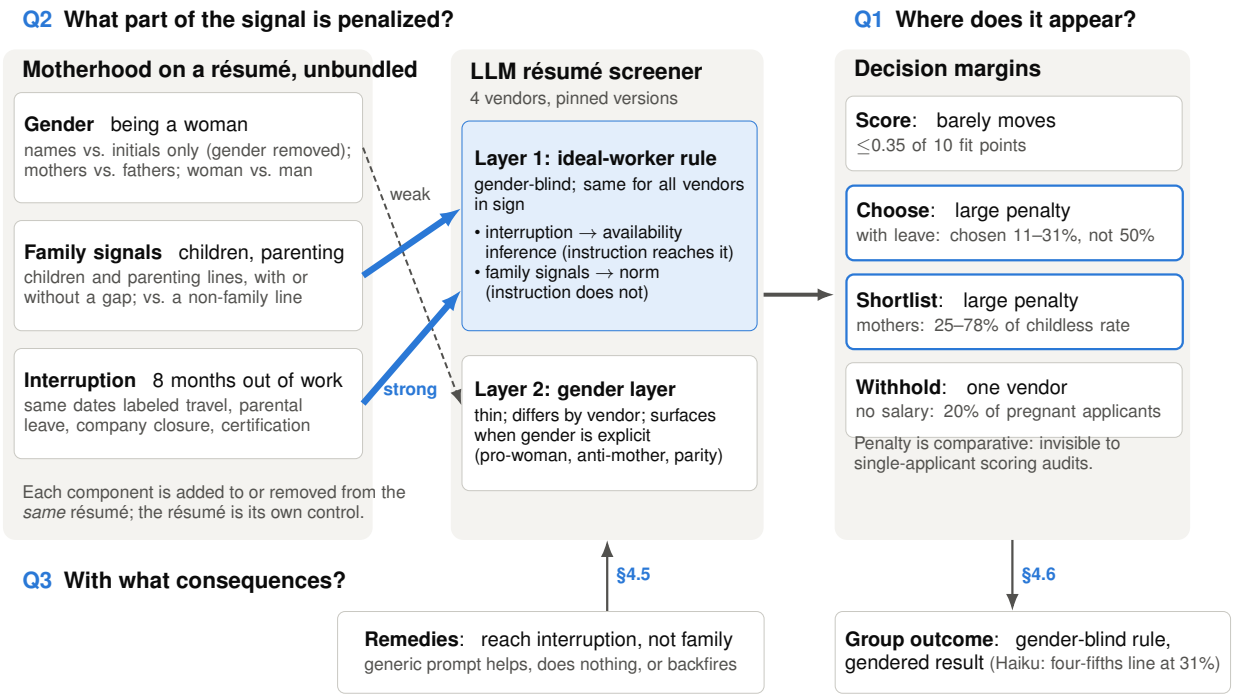


Figure 9: The framework revisited with the findings in place: a gender-blind caregiver penalty (Layer 1) carried by the interruption and family components, a thin vendor-specific gender layer (Layer 2), visible at the comparative margins.

A plausible reading, which our design cannot test directly, is that alignment training reshapes the behaviors it targets—overt differential treatment by gender—and leaves untouched the correlated judgments it does not target. Two models' explanations are suggestive: they declare family information irrelevant precisely in the decisions in which the parent loses. Without unaligned versions of these models, this remains a conjecture.

5.2 Gender-neutral in form, gendered in effect

The first layer is formally gender-neutral, but that does not make it neutral for gender equality. Caregiving interruptions and visible family responsibilities are carried disproportionately by women (Goldin 2014), so a rule that penalizes them reproduces a gender gap through the distribution of signals rather than the treatment of individuals—the structure that disparate-impact doctrine was designed to capture, and the long-standing argument that the ideal-worker norm disadvantages women without naming them (Acker 1990; Williams 2000). What is new here is the norm in its most abstract form: applied by systems that can read gender, have been trained not to use it, and penalize caregiving anyway.

The finding speaks to a broader question about how discrimination responds to its own regulation. When a protected category is policed, decisions can shift toward correlated signals that are not protected as such, as research on bans of criminal-record questions has shown (Agan and Starr 2018). The present case differs—the models are not substituting a proxy for a gender judgment they would otherwise make—but the consequence is similar: scrutiny organized around the protected category leaves the penalty on caregivers in place.

5.3 Implications for auditing

The framework also explains why standard audits would miss these patterns. Audits must be *comparative*, because the penalty lives in choices and shortlists, not in scores. They must *count non-answers*, because the largest pregnancy effect is a refusal to price that carries a denial of interviews. They should *measure gender knowledge implicitly*, because direct questions measure a model's willingness to state gender, not its ability to read it. And they must *look beyond protected categories*: a four-fifths comparison of women and men will miss a penalty that is gender-neutral in form, so audits need to compare applicants with and without caregiving signals. Because vendors differ widely—the same leave signal costs about twice as much choice probability with Haiku as with DeepSeek—an audit of one model says little about another, and the remedies phase cautions against relying on instructions: a generic fairness prompt moved the penalty in different directions for different vendors, and a targeted one removed only the component it named.

5.4 Limitations

The family signals are explicit, and real résumés carry weaker ones; the parental-leave gap is the most realistic. The materials are US-oriented, in English, text-only, and hold race constant, so the study is silent on how caregiver penalties intersect with race or class. Pregnancy has no male counterpart. Salary and interview outcomes are recommendations, not offers or callbacks. The models are four low-cost versions at fixed dates, used without system prompts except in the remedies phase; larger models and commercial products may behave differently, and versions change. Pools and pairs are constructed, so the design identifies the effect of a signal, not hiring rates. Group outcomes rest on scenario values for signal prevalence. The coding of explanations relies on LLM coders without human validation. Effective sample sizes are set by résumé pairs (40 or 16 per comparison), not by the number of calls.

6 Conclusion

Asked whether LLM résumé screeners penalize mothers, our answer is that the models we studied penalize caregivers. Unbundling the motherhood signal shows that they penalize time out of work and visible family responsibility, regardless of the applicant's gender, and that they do so only when applicants are compared. On this the four vendors agree. What differs among them is a thin gender layer and one vendor's practice of withholding decisions about pregnant applicants. Because the caregiver penalty is gender-neutral in form, it can pass fairness checks organized around gender while reproducing the motherhood penalty in effect.

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Items marked ⋄ are to be checked against the source before submission; items marked † were added in the manuscript and are not yet in the report's verified list. —

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